

Important Questions for Class 12

Maths

Chapter 5 – Continuity and Differentiability

Very Short Answer Questions

1 Mark

1. For what value of x , $f(x) = |2x - 7|$ is not derivable.

Ans: We need to find the input value for which the function will not be derivable. In other words, the point where the function has a break (there are two different tangents at that point)

$$f(x) = \begin{cases} 2x - 7 & x \geq \frac{7}{2} \\ 7 - 2x & x < \frac{7}{2} \end{cases}$$

Hence, for the value $x = \frac{7}{2}$, the function not be derivable.

2. Write the set of points of continuity of $g(x) = |x-1| + |x+1|$.

Ans: Let us check for the break points of the piecewise function,

$$g(x) = \begin{cases} (x-1) + (x+1) & x > 1 \\ (1-x) + (x+1) & -1 \leq x \leq 1 \\ (1-x) - (x+1) & x < -1 \end{cases}$$

We see that the function is a polynomial at all the breakpoints, polynomials are continuous for their entire domain and hence the points of continuity of $g(x)$ is $\mathbb{R}$

3. What is derivative of $|x-3|$ at $x=-1$

Ans: Writing the function in piecewise form, we get

$$f(x) = \begin{cases} x - 3 & x > 3 \\ 3 - x & x \leq 3 \end{cases}$$

For $x = -1$, we have

$$f(-1) = 3 - (-1)$$

$$= 4$$

$$f'(x) = \frac{d}{dx}(3 - x)$$

$$= -1$$

$$f'(-1) = -1$$

4. What are the points of discontinuity of $f(x) = \frac{(x-1)+(x+1)}{(x-7)(x-6)}$.

Ans: From the given function, it is noticeable that it is not defined for the values $x = 6, 7$. Considering that the function is a polynomial, we can say that it is continuous for the points mentioned above. Hence, the points of discontinuity will be situated at the inputs $x = 6$ and $x = 7$.

5. Write the number of points of discontinuity of $f(x) = [x]$ in $[3, 7]$.

Ans: We have that step functions are discontinuous at the integer points. At the given domain, the integer inputs are 4, 5, 6, 7. Hence, the number of points of discontinuity is 4.

6. The function, $f(x) = \begin{cases} \lambda x - 3 & x < 2 \\ 4 & x = 2 \\ 2x & x > 2 \end{cases}$ is a continuous function for all $x \in \mathbb{R}$,

find λ .

Ans: In piecewise functions, the function is continuous at the break points if the limit of the breakpoints evaluates to the same value. Evaluating the limit at the break points $x < 2$ and $x = 2$.

$$\lim_{x \rightarrow 2^-} \lambda x - 3 = \lim_{x \rightarrow 2} 4$$

$$\Rightarrow \lambda(2) - 3 = 4$$

$$\Rightarrow \lambda = \frac{7}{2}$$

Therefore, $\lambda = \frac{7}{2}$.

7. For what value of K, $f(x) = \begin{cases} \frac{\tan 3x}{\sin 2x} & x \neq 0 \\ 2K & x = 0 \end{cases}$ is continuous $\forall x \in \mathbb{R}$.

Ans: Evaluating the limit at the break points, we get

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} = 2K$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\frac{\tan 3x}{3x} \cdot 3}{\frac{\sin 2x}{2x} \cdot 2} = 2K$$

$$\Rightarrow \frac{3}{2} = 2K$$

$$\Rightarrow K = \frac{3}{4}$$

8. Write the derivative of $\sin x$ with respect to $\cos x$.

Ans: Let $y = \sin x$, and $z = \cos x$

Differentiate both with respect to x ,

$$\frac{dy}{dx} = \cos x \text{ and } \frac{dz}{dx} = -\sin x$$

Let us divide the first equation with the second equation, as shown below

$$\Rightarrow \frac{dy}{dx} \cdot \frac{dx}{dz} = \cos x \cdot \frac{1}{-\sin x}$$

$$\Rightarrow \frac{dy}{dz} = -\frac{\cos x}{\sin x}$$

$$\Rightarrow \frac{dy}{dz} = -\cot x$$

9. If $f(x) = x^2 g(x)$ and $g(1) = 6, g'(1) = 3$ find the value of $f'(1)$.

Ans: Differentiating the given function with respect to x ,

$$\frac{d}{dx}(f(x)) = 2xg(x) + g'(x)x^2$$

Therefore, $f'(1)$ should be,

$$f'(1) = 2(1)(6) + (3)(1)^2$$

$$= 12 + 3$$

$$= 15$$

10. Write the derivative of the following functions:

(i) $\log_3(3x+5)$

Ans: $y = \log_3(3x + 5)$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\log_3(3x + 5))$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{(3x + 5)\ln(3)}$$

(ii) $e^{\log_2 x}$

Ans: $y = e^{\log_2(x)}$

$$\Rightarrow \ln y = \log_2(x)$$

Differentiating with respect to x ,

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{y} \right) = \frac{d}{dx}(\log_2(x))$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{y} \right) = \frac{1}{(x)\ln(2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{\log_2(x)}}{x \ln(2)}$$

(iii) $e^{6\log_e(x-1)}, x > 1$

Ans:

$$y = e^{6 \log_e(x-1)}$$

$$\Rightarrow y = e^{\log_e(x-1)^6}$$

$$\Rightarrow y = (x-1)^6$$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}((x-1)^6)$$

$$\Rightarrow \frac{dy}{dx} = 6(x-1)^5$$

(iv) $\sec^{-1}\sqrt{x} + \csc^{-1}\sqrt{x}$, $x \geq 1$

Ans: $y = \sec^{-1}\sqrt{x} + \csc^{-1}\sqrt{x}$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\sec^{-1}\sqrt{x} + \csc^{-1}\sqrt{x})$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{(\sqrt{x})\sqrt{(\sqrt{x})^2 - 1}} + \frac{-1}{(\sqrt{x})\sqrt{(\sqrt{x})^2 - 1}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{x(x-1)}} - \frac{1}{\sqrt{x(x-1)}}$$

$$\Rightarrow \frac{dy}{dx} = 0$$

(v) $\sin^{-1}(x^{7/2})$

Ans: $y = \sin^{-1}(x^{7/2})$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx}(y) = \frac{d}{dx}(\sin^{-1}(x^{7/2}))$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-(x^{7/2})^2}} \cdot \left(\frac{7}{2}(x)^{\frac{5}{2}}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{7}{2} \cdot \frac{\sqrt{x^5}}{\sqrt{1-x^7}}$$

(vi) $\log_x 5, x > 0$

Ans: $y = \log_x (5)$

$$\Rightarrow y = \frac{\ln 5}{\ln x}$$

$$\Rightarrow \frac{1}{y} = \frac{\ln x}{\ln 5}$$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx} \left(\frac{1}{y} \right) = \frac{d}{dx} \left(\frac{\ln x}{\ln 5} \right)$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{y} \right) = \frac{1}{x \ln 5}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log_x 5}{x \ln 5}$$

Short Answer Questions

4 Marks

11. Discuss the continuity of following functions at the indicated points.

$$(i) f(x) = \begin{cases} \frac{x-|x|}{x} & x \neq 0 \\ 2 & x = 0 \end{cases} \text{ at } x=0$$

$$\text{Ans: Given, } f(x) = \begin{cases} \frac{x-|x|}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

Check the limit of the function as it approaches both sides of the point $x = 0$ from either direction,

Left-Hand limit is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x - (-x)}{x}$$

$$= \lim_{x \rightarrow 0^-} \frac{2x}{x}$$

$$= 2$$

Middle Value is,

$$f(0) = 2$$

Right-Hand limit is,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x - (x)}{x}$$

$$= 0$$

As all three limits do not evaluate to the same value, the function is discontinuous at the point $x = 0$.

$$(ii) \ g(x) = \begin{cases} \frac{\sin 2x}{3x} & x \neq 0 \\ \frac{3}{2} & x = 0 \end{cases} \quad \text{at } x=0$$

$$\text{Ans: Given, } g(x) = \begin{cases} \frac{\sin 2x}{3x} & x \neq 0 \\ \frac{3}{2} & x = 0 \end{cases}$$

Check the limit of the function as it approaches both sides of the point $x = 0$ from either direction.

Left-Hand limit is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\sin 2x}{3x}$$

$$= \lim_{x \rightarrow 0^-} \frac{\frac{\sin 2x}{3x} \cdot 2}{x}$$

$$= \frac{2}{3}$$

Middle Value is,

$$f(0) = \frac{3}{2}$$

Right-Hand limit is,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sin 2x}{3x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{\sin 2x}{3x} \cdot 2}{x}$$

$$= \frac{2}{3}$$

As all three limits do not evaluate to the same value, the function is discontinuous at the point $x = 0$.

$$(iii) \quad f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \quad \text{at } x=0$$

$$\text{Ans: Given, } f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Check the limit of the function as it approaches both sides of the point $x = 0$ from either direction.

Left-Hand limit is,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 \cos \frac{1}{x} \\ = 0$$

Middle Value is,

$$f(0) = 0$$

Right-Hand limit is,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^2 \cos \frac{1}{x} \\ = 0$$

We have that $\lim_{x \rightarrow 0^-} f(x) = f(0) = \lim_{x \rightarrow 0^+} f(x)$

As all three limits evaluate to the same value, the function is continuous at the point $x = 0$.

(iv) $f(x) = |x| + |x-1|$ at $x=1$

Ans: Given, $f(x) = |x| + |x-1|$

Check the limit of the function as it approaches both sides of the point $x = 1$ from either direction.

Left-Hand limit is,

$$\lim_{x \rightarrow 1^-} (|x| + |x-1|) = \lim_{x \rightarrow 1^-} (x + (1-x)) \\ = 1$$

Middle Value is,

$$f(1) = (1) + (1-1)$$

$$= 1$$

Right-Hand limit is,

$$\lim_{x \rightarrow 1^+} (|x| + |x-1|) = \lim_{x \rightarrow 1^+} (x + (x-1)) \\ = 1$$

We have that $\lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$

As all three limits evaluate to the same value, the function is continuous at the point $x = 1$.

$$(v) f(x) = \begin{cases} x - [x] & x \neq 1 \\ 0 & x = 1 \end{cases} \text{ at } x=1$$

$$\text{Ans: Given, } f(x) = \begin{cases} x - [x] & x \neq 1 \\ 0 & x = 1 \end{cases}$$

Check the limit of the function as it approaches both sides of the point $x = 1$ from either direction.

Left-Hand limit is,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x - [x])$$

$$= 1 - (0)$$

$$= 1$$

Middle Value is,

$$f(1) = (1) - (1)$$

$$= 0$$

Right-Hand limit is,

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x - [x])$$

$$= 1 - (1)$$

$$= 0$$

As all three limits do not evaluate to the same value, the function is discontinuous at the point $x = 1$.

$$12. \text{ For what value of } k, f(x) = \begin{cases} 3x^2 - kx + 5 & 0 \leq x < 2 \\ 1 - 3x & 2 \leq x \leq 3 \end{cases} \text{ is continuous } \forall x \in [0, 3].$$

Ans: If the function is continuous, the limit at the break points must evaluate to the same value.

Therefore, the following condition must be met,

$$\lim_{x \rightarrow 2^-} 3x^2 - kx + 5 = \lim_{x \rightarrow 2^+} 1 - 3x$$

$$\Rightarrow 3(2)^2 - k(2) + 5 = 1 - 3(2)$$

$$\Rightarrow 12 - 2k + 5 = -5$$

$$\Rightarrow 2 = 2k$$

$$\Rightarrow k = 1$$

Secondly, since the piecewise function consists of polynomials, we know that the polynomials are continuous for their respective domains. Hence, the value of k must be $k = 1$ if the function is continuous $\forall x \in [0, 3]$.

13. For what values of a and b , $f(x) = \begin{cases} \frac{x+2}{|x+2|} + a & \text{if } x < -2 \\ a+b & \text{if } x = -2 \\ \frac{x+2}{|x+2|} + 2b & \text{if } x > -2 \end{cases}$ is continuous at $x = -2$

Ans: For the function to be continuous, the following condition has to be necessarily met,

$$\lim_{x \rightarrow -2^-} \left(\frac{x+2}{|x+2|} + a \right) = a + b = \lim_{x \rightarrow -2^+} \left(\frac{x+2}{|x+2|} + 2b \right)$$

Left-Hand limit is,

$$\Rightarrow \lim_{x \rightarrow -2^-} \left(\frac{x+2}{|x+2|} + a \right) = a + b$$

$$\Rightarrow \lim_{x \rightarrow -2^-} \left(\frac{x+2}{-(x+2)} + a \right) = a + b$$

$$\Rightarrow -1 + a = a + b$$

$$\Rightarrow b = -1$$

Right-Hand limit is,

$$\Rightarrow a + b = \lim_{x \rightarrow -2^+} \left(\frac{x+2}{|x+2|} + 2b \right)$$

$$\Rightarrow a + b = \lim_{x \rightarrow -2^+} \left(\frac{x+2}{x+2} + 2b \right)$$

$$\Rightarrow a + b = 1 + 2b$$

$$\Rightarrow a - 1 = 1 - 2$$

$$\Rightarrow a = 0$$

The values of a and b should be $a = 0$ and $b = -1$.

14. Prove that $f(x) = |x+1|$ is continuous at $x = -1$ but not derivable at $x = -1$.

Ans: To prove that the function is continuous at $x = -1$, take the limit of the breaking points,

$$\begin{aligned}\lim_{x \rightarrow -1^-} |x+1| &= |-1+1| = \lim_{x \rightarrow -1^+} |x+1| \\ \Rightarrow \lim_{x \rightarrow -1^-} -(x+1) &= |0| = \lim_{x \rightarrow -1^+} (x+1) \\ \Rightarrow 0 &= 0 = 0\end{aligned}$$

Since the above condition is satisfied, we conclude that the function is continuous.

We have proved continuity, now we need to check for the left-hand derivative and the right-hand derivative.

$$\begin{aligned}\Rightarrow \lim_{h \rightarrow 0^+} \frac{f(-1+h) - f(-1)}{h} &= \lim_{h \rightarrow 0^+} \frac{f(-1+h) - f(-1)}{h} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{|(-1+h)+1| - |-1+1|}{h} &= \lim_{h \rightarrow 0^+} \frac{|(-1+h)+1| - |-1+1|}{h} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{(-1+h+1)}{h} &= \lim_{h \rightarrow 0^+} \frac{-(-1+h+1)}{h} \\ \Rightarrow 1 &\neq -1\end{aligned}$$

Since the above condition could not be met, the function is not differentiable or derivable at $x = -1$.

15. For what value of p, $f(x) = \begin{cases} x^p \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is derivable at $x = 0$

Ans: To ensure derivability, we must ensure the following condition

$$\begin{aligned}\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{(0+h)^p \sin\left(\frac{1}{0+h}\right) - 0}{h} &= \lim_{h \rightarrow 0^+} \frac{(0+h)^p \sin\left(\frac{1}{0+h}\right) - 0}{h}\end{aligned}$$

$$\begin{aligned}\Rightarrow \lim_{h \rightarrow 0^+} \frac{h^p \sin\left(\frac{1}{h}\right)}{h} &= \lim_{h \rightarrow 0^-} \frac{h^p \sin\left(\frac{1}{h}\right)}{h} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{h^{p-1} \sin\left(\frac{1}{h}\right)}{h \cdot \frac{1}{h}} &= \lim_{h \rightarrow 0^-} \frac{h^{p-1} \sin\left(\frac{1}{h}\right)}{h \cdot \frac{1}{h}} \\ \Rightarrow \lim_{h \rightarrow 0^+} \frac{h^{p-1}}{h} &= \lim_{h \rightarrow 0^-} \frac{h^{p-1}}{h}\end{aligned}$$

Therefore, the condition that we have obtained for p is as shown below,
 $p-1 > 0 \Rightarrow p > 1$ which means p must be greater than 1.

16. If $y = \frac{1}{2} \left[\tan^{-1}\left(\frac{2x}{1-x^2}\right) + 2\tan^{-1}\left(\frac{1}{x}\right) \right]$, $0 < x < 1$, find $\frac{dy}{dx}$.

Ans: Differentiating with respect to x ,

$$\begin{aligned}\Rightarrow \frac{d}{dx}(y) &= \frac{1}{2} \frac{d}{dx} \left(\tan^{-1}\left(\frac{2x}{1-x^2}\right) \right) + \frac{d}{dx} \left(\tan^{-1}\left(\frac{1}{x}\right) \right) \\ \Rightarrow \frac{dy}{dx} &= \frac{1}{2 \left(1 + \left(\frac{2x}{1-x^2} \right)^2 \right)} \cdot \frac{2(1-x^2) - 2x(-2x)}{(1-x^2)^2} + \frac{1}{1 + \left(\frac{1}{x} \right)^2} \cdot \left(-\frac{1}{x^2} \right) \\ \Rightarrow \frac{dy}{dx} &= \frac{2(1-x^2) + 4x^2}{2((1-x^2)^2 + 4x^2)} - \frac{1}{x^2 + 1} \\ \Rightarrow \frac{dy}{dx} &= \frac{(1+x^2)}{(1+x^2)^2} - \frac{1}{x^2 + 1} \\ \Rightarrow \frac{dy}{dx} &= 0\end{aligned}$$

17. If $y = \sin \left[2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right]$ then $\frac{dy}{dx} = ?$

Ans: Let $x = \cos t$

We have,

$$\Rightarrow y = \sin \left[2 \tan^{-1} \sqrt{\frac{1 - \cos t}{1 + \cos t}} \right]$$

$$\Rightarrow y = \sin \left[2 \tan^{-1} \sqrt{\tan \left(\frac{t}{2} \right)} \right]$$

$$\Rightarrow y = \sin \left[2 \left(\frac{t}{2} \right) \right]$$

$$\Rightarrow y = \sin t$$

$$\Rightarrow y = \sqrt{1 - \cos^2 t}$$

$$\Rightarrow y = \sqrt{1 - x^2}$$

Differentiating with respect to x ,

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} (1 - x^2)^{-\frac{1}{2}} (-2x)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{\sqrt{1 - x^2}}$$

18. If $5^x + 5^y = 5^{x+y}$ then prove that $\frac{dy}{dx} + 5^{y-x} = 0$.

Ans: Differentiating with respect to x , we get

$$5^x \ln 5 + 5^y \ln 5 \frac{dy}{dx} = 5^{x+y} \ln 5 \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow 5^x + \frac{dy}{dx} 5^y = \left(1 + \frac{dy}{dx} \right) 5^{x+y}$$

$$\Rightarrow 5^x + \frac{dy}{dx} 5^y = 5^{x+y} + \frac{dy}{dx} 5^{x+y}$$

$$\Rightarrow \frac{dy}{dx}(5^{x+y} - 5^y) + (5^{x+y} - 5^x) = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{5^{x+y} - 5^x}{5^{x+y} - 5^y} = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{5^y}{5^x} = 0$$

$$\Rightarrow \frac{dy}{dx} + 5^{y-x} = 0$$

19. If $x\sqrt{1-y^2} + y\sqrt{1-x^2} = a$ then show that $\frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$.

Ans: Differentiating with respect to x

$$\Rightarrow -\frac{2xy}{\sqrt{1-y^2}} \frac{dy}{dx} + \sqrt{1-y^2} - \frac{2xy}{\sqrt{1-x^2}} + \frac{dy}{dx} \sqrt{1-x^2} = 0$$

$$\Rightarrow \frac{dy}{dx} \left(\sqrt{1-x^2} - \frac{2xy}{\sqrt{1-y^2}} \right) = \frac{2xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\frac{2xy - (\sqrt{1-y^2}\sqrt{1-x^2})}{\sqrt{1-y^2}}}{\frac{2xy - (\sqrt{1-y^2}\sqrt{1-x^2})}{\sqrt{1-x^2}}}$$

$$\Rightarrow \frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$

20. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then show that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.

Ans: Let,

$$x = \sin m$$

$$m = \sin^{-1} x$$

$$y = \sin n$$

$$n = \sin^{-1} y$$

We have,

$$\Rightarrow \sqrt{1 - \sin^2 m} + \sqrt{1 - \sin^2 n} = a(\sin m - \sin n)$$

$$\Rightarrow \cos m + \cos n = a \left(2 \cos \frac{m+n}{2} \sin \frac{m-n}{2} \right)$$

$$\Rightarrow 2 \cos \frac{m+n}{2} \cos \frac{m-n}{2} = 2a \cos \frac{m+n}{2} \cos \frac{m-n}{2}$$

$$\Rightarrow \frac{1}{a} = \tan \frac{m-n}{2}$$

$$\Rightarrow \tan^{-1} \frac{1}{a} = \frac{m-n}{2}$$

$$\Rightarrow 2 \tan^{-1} \frac{1}{a} = \sin^{-1} x - \sin^{-1} y$$

Differentiating with respect to x ,

$$\Rightarrow 0 = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$

21. If $(x+y)^{m+n} = x^m y^n$ then prove that $\frac{dy}{dx} = \frac{y}{x}$.

Ans: Take the natural log on both sides as shown below,

$$\ln((x+y)^{m+n}) = \ln(x^m y^n)$$

$$\Rightarrow (m+n) \ln(x+y) = m \ln x + n \ln y$$

Differentiating with respect to x ,

$$\Rightarrow \frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{m+n}{x+y} \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{n}{y} - \frac{m+n}{x+y} \right) = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{nx + ny - my - ny}{y(x+y)} \right) = \frac{mx + nx - mx - my}{x(x+y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

22. Find the derivative of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ w.r.t. $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$.

Ans: $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{2x}{1-x^2}\right)^2} \cdot \frac{2(1-x^2) - 2x(-2x)}{(1-x^2)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2}$$

Let $x = \tan t$.

$$z = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

$$\Rightarrow z = \sin^{-1}(\sin 2t)$$

$$\Rightarrow z = 2 \tan^{-1} x$$

$$\Rightarrow \frac{dz}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| < 1 \\ -\frac{2}{1+x^2} & |x| \geq 1 \end{cases}$$

Hence, we get,

$$\frac{dy}{dz} = \begin{cases} 1 & |x| < 1 \\ -1 & |x| \geq 1 \end{cases}$$

23. Find the derivative of $\ln(\sin x)$ w.r.t. $\log_a(\cos x)$.

Ans: Let,

$$y = \ln(\sin x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sin x} \cos x$$

$$z = \log_a \cos x$$

$$\Rightarrow z = \frac{\ln \cos x}{\ln a}$$

$$\Rightarrow \frac{dz}{dx} = \frac{1(-\sin x)}{\cos x} \cdot \frac{1}{\ln a}$$

Hence, we get

$$\begin{aligned} \frac{dy}{dx} \cdot \frac{dx}{dz} &= \frac{\cos x}{\sin x} \cdot \frac{\ln a \cos x}{-\sin x} \\ &= -\ln a \cot^2 x \end{aligned}$$

24. If $x^y + y^x + x^x = m^n$, then find the value of $\frac{dy}{dx}$.

Ans: On differentiating with respect to x , we get

$$\Rightarrow \frac{d}{dx}(x^y) + \frac{d}{dx}(y^x) + \frac{d}{dx}(x^x) = 0$$

$$\Rightarrow x^y \left(\frac{dy}{dx} \ln x + \frac{y}{x} \right) + y^x \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) + x^x (\ln x + 1) = 0$$

Now, further simplify the expression obtained in the equation above as shown below,

$$\Rightarrow \frac{dy}{dx} x^y \ln x + y^x \frac{x}{y} \frac{dy}{dx} = -x^x (\ln x + 1) - y^x \ln y - x^y \left(\frac{y}{x} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x^x (\ln x + 1) - y^x \ln y - x^{y-1} y}{y^{x-1} x + x^y \ln x}$$

25. If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ then find $\frac{d^2 y}{dx^2}$.

Ans: Differentiate with respect to θ ,

$$\Rightarrow \frac{d}{d\theta}(x) = \frac{d}{d\theta}(a \cos^3 \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

Also,

$$\Rightarrow \frac{d}{d\theta}(y) = \frac{d}{d\theta}(a \sin^3 \theta)$$

$$\Rightarrow \frac{dy}{d\theta} = 3a \sin^2 \theta (\cos \theta)$$

Therefore, we have

$$\frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = -\frac{3a \sin^2 \theta \cos \theta}{3a \cos^2 \theta \sin \theta}$$

$$\Rightarrow \frac{dy}{dx} = -\tan \theta$$

Differentiating with respect to x ,

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (-\tan \theta)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{d}{d\theta} (-\tan \theta) \frac{d\theta}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\sec^2 \theta \left(-\frac{1}{3a \cos^2 \theta \sin \theta} \right)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{1}{3a \sin \theta \cos^4 \theta}$$

26. If $x = ae^t(\sin t - \cos t)$, $y = ae^t(\sin t + \cos t)$ then show that $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is 1.

Ans: Differentiate the first equation with respect to t

$$\Rightarrow \frac{d}{dt}(x) = \frac{d}{dt}(ae^t(\sin t - \cos t))$$

$$\Rightarrow \frac{dx}{dt} = ae^t(\sin t - \cos t) + ae^t(\cos t + \sin t)$$

$$\Rightarrow \frac{dx}{dt} = 2ae^t \sin t$$

Differentiate the second equation with respect to t

$$\Rightarrow \frac{d}{dt}(y) = \frac{d}{dt}(ae^t(\sin t + \cos t))$$

$$\Rightarrow \frac{dy}{dt} = ae^t(\sin t + \cos t) + ae^t(\cos t - \sin t)$$

$$\Rightarrow \frac{dy}{dt} = 2ae^t \cos t$$

Therefore, we have

$$\frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{2ae^t(\cos t)}{2ae^t(\sin t)}$$

$$\Rightarrow \frac{dy}{dx} = \cos t$$

Therefore at $t = \frac{\pi}{4}$, we have

$$\begin{aligned} \frac{dy}{dx} &= \cot\left(\frac{\pi}{4}\right) \\ &= 1 \end{aligned}$$

27. If $y = \sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$ then find $-\frac{dy}{dx}$.

Ans: We have,

$$y = \sin^{-1}\left[x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2}\right]$$

Using the inverse sine addition trigonometric identity

$$[\sin^{-1} x - \sin^{-1} y = \sin^{-1}[x\sqrt{1-y^2} - y\sqrt{1-x^2}]], \text{ we have}$$

$$y = \sin^{-1} x - \sin^{-1} \sqrt{x}$$

Differentiating with respect to x

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$$

$$\Rightarrow -\frac{dy}{dx} = \frac{1}{2\sqrt{x(1-x)}} - \frac{1}{\sqrt{1-x^2}}$$

28. If $y = x^{\ln x} + (\ln x)^x$ then find $\frac{dy}{dx}$.

Ans: Let,

$$y = u + v \text{ where } u = x^{\ln x} \text{ and } v = (\ln x)^x$$

Differentiating with respect to x

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

Now, for u

$$u = x^{\ln x}$$

Take the natural logarithm of both sides

$$\Rightarrow \ln u = \ln x \ln x$$

$$\Rightarrow \ln u = (\ln x)^2$$

Differentiating with respect to x

$$\Rightarrow \frac{du}{dx} \cdot \frac{1}{u} = 2 \ln x \left(\frac{1}{x} \right)$$

$$\Rightarrow \frac{du}{dx} = \frac{2x^{\ln x} \ln x}{x}$$

Now, for v

$$v = (\ln x)^x$$

Take the natural logarithm of both sides

$$\ln v = x \ln(\ln x)$$

Differentiating with respect to x

$$\Rightarrow \frac{dv}{dx} \cdot \frac{1}{v} = x \left[\frac{1}{\ln x} \cdot \frac{1}{x} \right] + \ln(\ln x)$$

$$\Rightarrow \frac{dv}{dx} = (\ln x)^x \left(\frac{1}{\ln x} + \ln(\ln x) \right)$$

Putting both back in the equation, we get

$$\frac{dy}{dx} = \frac{2x^{\ln x} \ln x}{x} + (\ln x)^x \left(\frac{1}{\ln x} + \ln(\ln x) \right)$$

29. Differentiate x^{x^x} w.r.t. x .

Ans: Taking the natural logarithm on both sides and then simplifying further, it can be obtained as show below,

$$y = x^{x^x}$$

$$\Rightarrow \ln y = \ln(x^{x^x})$$

$$\Rightarrow \ln y = x^x \ln x$$

Differentiating with respect to x ,

$$\Rightarrow \frac{dy}{dx} \frac{1}{y} = \frac{x^x}{x} + \ln x (x^x (1 + \ln x))$$

$$\Rightarrow \frac{dy}{dx} = x^{x^x} (x^{x-1} + \ln x (x^x (1 + \ln x)))$$

30. Find $\frac{dy}{dx}$, if $(\cos x)^y = (\cos y)^x$.

Ans: Taking the natural logarithm on both sides

$$\Rightarrow \ln((\cos x)^y) = \ln((\cos y)^x)$$

$$\Rightarrow y \ln \cos x = x \ln \cos y$$

Differentiating both sides with respect to x

$$\Rightarrow \frac{d}{dx} (y \ln(\cos x)) = \frac{d}{dx} (x \ln(\cos y))$$

$$\Rightarrow y \left(\frac{1}{\cos x} (-\sin x) \right) + \frac{dy}{dx} \ln \cos x = x \left(\frac{1}{\cos y} (-\sin y) \left(\frac{dy}{dx} \right) \right) + \ln \cos y$$

$$\Rightarrow \frac{dy}{dx}(\ln \cos x) - y \tan x = \frac{dy}{dx}(-x \tan y) + \ln \cos y$$

$$\Rightarrow \frac{dy}{dx}(\ln \cos x + x \tan y) = y \tan x + \ln \cos y$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \tan x + \ln \cos y}{x \tan y + \ln \cos x}$$

31. If $y = \tan^{-1} \left(\frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\sqrt{1+\sin x} + \sqrt{1-\sin x}} \right)$ where $\frac{\pi}{2} < x < \pi$, find $\frac{dy}{dx}$.

Ans: We have

$$y = \tan^{-1} \left(\frac{\sqrt{1-\sin x} - \sqrt{1+\sin x}}{\sqrt{1-\sin x} + \sqrt{1+\sin x}} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{\sqrt{1-\sin x} - \sqrt{1+\sin x}}{\sqrt{1-\sin x} + \sqrt{1+\sin x}} \times \frac{\sqrt{1-\sin x} - \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{(\sqrt{1-\sin x} - \sqrt{1+\sin x})^2}{(1-\sin x) - (1+\sin x)} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{(\sqrt{1-\sin x})^2 + (\sqrt{1+\sin x})^2 - 2\sqrt{1-\sin x}\sqrt{1+\sin x}}{-2\sin x} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{2 - 2\sqrt{\cos^2 x}}{-2\sin x} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{1 - \cos x}{-\sin x} \right)$$

$$\Rightarrow y = \tan^{-1} \left(\frac{2\sin^2 \frac{x}{2}}{-2\sin \frac{x}{2} \cos \frac{x}{2}} \right)$$

$$\Rightarrow y = \tan^{-1}\left(-\tan \frac{x}{2}\right)$$

$$\Rightarrow y = -\tan^{-1}\left(\tan \frac{x}{2}\right)$$

$$\Rightarrow y = -\frac{x}{2}$$

Differentiating with respect to x

$$\frac{dy}{dx} = -\frac{1}{2}$$

32. If $x = \sin\left(\frac{1}{a} \ln y\right)$, then show that $(1-x^2)y'' - xy' - a^2y = 0$.

Ans: We have

$$x = \sin\left(\frac{1}{a} \ln y\right)$$

$$\Rightarrow \sin^{-1} x = \frac{1}{a} \ln y$$

$$\Rightarrow a \sin^{-1} x = \ln y$$

$$\Rightarrow e^{a \sin^{-1} x} = y$$

Differentiating with respect to x

$$\Rightarrow \frac{dy}{dx} = e^{a \sin^{-1} x} \left(\frac{a}{\sqrt{1-x^2}} \right)$$

$$\Rightarrow y' = \frac{ay}{\sqrt{1-x^2}}$$

Differentiating with respect to x

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{a \frac{dy}{dx} \sqrt{1-x^2} + \frac{xay}{\sqrt{1-x^2}}}{(1-x^2)^2}$$

$$\Rightarrow y'' = \frac{ay'(1-x^2) + xay}{(1-x^2)\sqrt{1-x^2}}$$

$$\Rightarrow y'' = \frac{ay'}{\sqrt{1-x^2}} + \frac{xay}{(1-x^2)\sqrt{1-x^2}}$$

$$\Rightarrow y'' = \frac{a^2y}{1-x^2} + \frac{xy'}{(1-x^2)}$$

$$\Rightarrow (1-x^2)y'' - xy' - a^2y = 0$$

33. Differentiate $(\ln x)^{\ln x}$, $x > 1$ w.r.t. x .

Ans: Taking the natural log on both sides

$$\ln y = \ln x (\ln(\ln x))$$

Differentiating with respect to x

$$\Rightarrow \frac{dy}{dx} \frac{1}{y} = \ln x \left(\frac{1}{x \ln x} \right) + \frac{\ln(\ln x)}{x}$$

$$\Rightarrow \frac{dy}{dx} = (\ln x)^{\ln x} \left(\frac{1 + \ln(\ln x)}{x} \right)$$

34. If $\sin y = x \sin(a+y)$ then show that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$.

Ans: We have

$$\sin y = x \sin(a+y)$$

$$\Rightarrow x = \frac{\sin y}{\sin(a+y)}$$

Differentiating with respect to y

$$\Rightarrow \frac{dx}{dy} = \frac{\cos y (\sin(a+y)) - \sin y (\cos(a+y))}{\sin^2(a+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\cos y (\sin a \cos y + \cos a \sin y) - \sin y (\cos a \cos y - \sin a \sin y)}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\cos^2 y \sin a + \cos a \sin y \cos y - \sin y \cos a \cos y + \sin a \sin^2 y}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\cos^2 y \sin a + \sin^2 y \sin a}{\sin^2 a + y}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a (\cos^2 y + \sin^2 y)}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a + y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a}$$

35. If $y = \sin^{-1}x$, find $\frac{d^2y}{dx^2}$ in terms of y .

Ans: Differentiating with respect to x

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Differentiating with respect to x

$$\frac{d^2y}{dx^2} = -\frac{1}{2}(1-x^2)^{-\frac{3}{2}}(-2x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{x}{(\sqrt{1-x^2})^3}$$

Now, we have

$$y = \sin^{-1}x$$

$$\Rightarrow x = \sin y$$

Replacing in our equation

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin y}{(\sqrt{1 - \sin^2 y})^3}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin y}{(\cos y)^3}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \tan y \sec^2 y$$

36. If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then show that $\frac{d^2y}{dx^2} = \frac{-b^4}{a^2 y^3}$.

Ans: Differentiating with respect to x

$$\Rightarrow \frac{2x}{a^2} + \frac{dy}{dx} \left(\frac{2y}{b^2} \right) = 0$$

$$\Rightarrow \frac{x}{a} = -\frac{dy}{dx} \left(\frac{y}{b^2} \right)$$

Differentiating with respect to x

$$\Rightarrow \frac{1}{a^2} = -\left(\left(\frac{dy}{dx} \right)^2 \frac{1}{b^2} + \frac{d^2y}{dx^2} \frac{y}{b^2} \right)$$

$$\Rightarrow \frac{1}{a^2} + \frac{(y')^2}{b^2} + \frac{yy''}{b^2} = 0$$

Now, we know

$$\frac{x}{a^2} = -\frac{yy'}{b^2}$$

$$\Rightarrow \frac{y'}{b^2} = -\frac{x}{a^2 y}$$

$$\Rightarrow \left(\frac{y'}{b^2} \right)^2 = \left(-\frac{x}{a^2 y} \right)^2$$

$$\Rightarrow \frac{(y')^2}{b^4} = \frac{x^2}{a^4 y^2}$$

$$\Rightarrow \frac{(y')^2}{b^2} = \frac{x^2 b^2}{a^4 y^2}$$

Substituting in our previous equation

$$\frac{1}{a^2} + \frac{x^2 b^2}{a^4 y^2} + \frac{yy''}{b^2} = 0$$

$$\Rightarrow \frac{1}{a^2} \left[1 + \frac{x^2}{a^2} \cdot \frac{b^2}{y^2} \right] + \frac{yy''}{b^2} = 0$$

$$\Rightarrow \frac{1}{a^2} \left[1 + \left(1 - \frac{y^2}{b^2} \right) \frac{b^2}{y^2} \right] + \frac{yy''}{b^2} = 0$$

$$\Rightarrow \frac{1}{a^2} \left[\frac{b^2}{y^2} \right] + \frac{yy''}{b^2} = 0$$

$$\Rightarrow \frac{yy''}{b^2} = -\frac{b^2}{a^2 y^2}$$

$$\Rightarrow y'' = -\frac{b^4}{a^2 y^3}$$

37. If $y = e^{a \cos^{-1} x}$, $-1 \leq x \leq 1$, show that $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$.

Ans: Differentiating with respect to x

$$\Rightarrow \frac{dy}{dx} = e^{a \cos^{-1} x} \left(-\frac{a}{\sqrt{1-x^2}} \right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{ay}{\sqrt{1-x^2}}$$

Differentiating with respect to x

$$\Rightarrow \frac{d^2 y}{dx^2} = -\frac{a \frac{dy}{dx} \sqrt{1-x^2} - \frac{xay}{\sqrt{1-x^2}}}{(1-x^2)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{a \frac{dy}{dx}(1-x^2) - xay}{(1-x^2)\sqrt{1-x^2}}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{a \frac{dy}{dx}}{\sqrt{1-x^2}} + \frac{xay}{(1-x^2)\sqrt{1-x^2}}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{a^2y}{1-x^2} + \frac{x \frac{dy}{dx}}{(1-x^2)}$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} + a^2y - x \frac{dy}{dx} = 0$$

38. If $y^3 = 3ax^2 - x^3$ then prove that $\frac{d^2y}{dx^2} = \frac{-2a^2x^2}{y^5}$.

Ans: Differentiating with respect to x

$$\Rightarrow 3y^2 \frac{dy}{dx} = 6ax - 3x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2ax}{y^2} - \frac{x^2}{y^2}$$

Differentiating with respect to x

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(2a - 2x)y^2 - (2ax - x^2)2y \left(\frac{dy}{dx} \right)}{y^4}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(2a - 2x)y^2 - (2ax - x^2)2y \left(\frac{2ax - x^2}{y^2} \right)}{y^4}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(2a - 2x)y^3 - 2(2ax - x^2)^2}{y^5}$$

$$\begin{aligned}\Rightarrow \frac{d^2y}{dx^2} &= \frac{(2a - 2x)(3ax^2 - x^3) - 2(4a^2x^2 + x^4 - 4ax^3)}{y^5} \\ \Rightarrow \frac{d^2y}{dx^2} &= \frac{6a^2x^2 - 6ax^3 - 2ax^3 + 2x^4 - 8a^2x^2 - 2x^4 + 8ax^3}{y^5} \\ \Rightarrow \frac{d^2y}{dx^2} &= -\frac{2a^2x^2}{y^5}\end{aligned}$$

39. Verify Rolle's theorem for the function, $y=x^2+2$ in the interval $[a,b]$ where $a=-2, b=2$.

Ans: We have been given a continuous function which is also differentiable in the given domain [since it is a polynomial].

We also have

$$f(a) = f(-2) = (-2)^2 + 2 = 6$$

$$f(b) = f(2) = (2)^2 + 2 = 6$$

Therefore, we have $f(a) = f(b)$

So, according to Rolle's theorem, there must exist a point c , such that $f'(c) = 0$

Let us find c

$$f'(x) = 2x$$

Replacing $f'(x) = 0$

$$0 = 2x$$

$$x = 0 = c$$

We know that $c \in [-2, 2]$

Hence, Rolle's theorem is verified for the given function.

40. Verify Mean Value Theorem for the function, $f(x)=x^2$ in $[2,4]$

Ans: We have been given a continuous function which is also differentiable in the given domain [since it is a polynomial].

The Mean Value Theorem states that there must exist a point c such that

$$f'(c) = \frac{f(4) - f(2)}{4 - 2}$$

We know that $f'(x) = 2x$, Let us find the point c

$$f'(c) = \frac{(4)^2 - (2)^2}{4 - 2}$$

$$\Rightarrow 2c = \frac{12}{2}$$

$$\Rightarrow 2c = 6$$

$$\Rightarrow c = 3$$

We know that $c \in [2, 4]$

Hence, Mean Value Theorem is verified for the given function.